QEO1 – Manuscripts in the QED3 construction

One box per manuscript and its companion outline, or per outline for a future manuscript. Every document occurs in one box only. Shared chamber setup and final parameter admission belong to QEM1. Arrows identify the results used, under their stated hypotheses.

MODEL-SPECIFIC Proof draft written

LPM1 / LPO1
Lattice laws and finite estimates

Gaussian, trace and chessboard bounds; infinite-lattice curvature laws with all nine RP families.

Proof manuscript and companion outline

Draft: 53 pages
Difficulty: 5/5

Finite traces and nine reflection families.

MODEL-SPECIFIC Proof draft written

MBM1 / MBO1
Maxwell block estimates

Constrained means, local bounds and the exact Gaussian split, including the stated weakening and domain estimates.

Proof manuscript and companion outline

Draft: 15 pages
Difficulty: 3/5

Uniform constrained Gaussian estimates.

MODEL-SPECIFIC Proof draft written

FBM1 / FBO1
Chamber fermion estimates

Free kernels, weighted local integration and determinant gains. Reconstructed connections use Maxwell bounds.

Proof manuscript and companion outline

Draft: 27 pages
Difficulty: 4/5

Chamber inverses and determinant gains.

MODEL-SPECIFIC Proof draft written

RNM1 / RNO1
Extraction and contraction

Relevant extraction, bounded reconstruction and normalized contraction, with their distinct estimates.

Proof manuscript and companion outline

Draft: 27 pages
Difficulty: 4/5

Reconstruction and normalized contraction.

all three

MODEL-SPECIFIC Proof draft written

NLM1 / NLO1
Exact density and nonlinear step

Construct the local density, bound its nonlinear remainder, and close one renormalization step.

Proof manuscript and companion outline; QEO1/S12–S15

Draft: 33 pages
Difficulty: 5/5

Localization and the connected logarithm.

free kernels

MODEL-SPECIFIC Proof draft written

PRM1 / PRO1
Positive probe and free response

Positive quartic coefficient and free coefficient comparisons for the translated-product argument.

Proof manuscript and companion outline written

Draft: 18 pages
Difficulty: 4/5

Quartic sign and uniform trace comparison.

select the same bare sequence

both

INTERFACE Proof draft written

CEM1
Continuum existence and inheritance

Gaussian and correlation stability, compactness, structural inheritance and prescribed-parent weak recovery.

CEO1 is its compactness outline. The finite-periodic moment application belongs to VTM1.

Draft: 15 pages
Difficulty: 2/5

Compactness, inheritance and recovery.

MODEL-SPECIFIC Proof draft written

CVM1 / CVO1
Covers and common tuning

Odd-cover agreement and one bare pair per cutoff, transported to the specified reference tori.

Proof manuscript and companion outline; QEO1/S16–S18, T01

Draft: 31 pages
Difficulty: 4/5

Exact cover agreement and common tuning.

MODEL-SPECIFIC Proof draft written

RRM1 / RRO1
Source identities and reference response
Early results: T05–T07.

Construct the sources and prove the zero-flux/harmonic and Wick identities. T06 is an input to the terminal measure.

T06: zero flux

T07: Wick identity

Later result: Q4R, T12–T15.

Using the terminal measure, control the normalized mixed response through degree four in each cell variable, including the complex complement and remainder.

Proof manuscript and companion outline written. Both parts belong to this one work.

Draft: 35 pages
Difficulty: 5/5

Marked derivatives and complex remainder.

INTERFACE Proof draft written

ECM1 / ECO1
Ergodic components and clustering

Inherited reflection forms and Gaussian bounds, with clustering on the same full-measure component set.

Proof manuscript and companion outline; QEO1/B2

Draft: 8 pages
Difficulty: 3/5

Conditional inheritance and clustering.

MODEL-SPECIFIC Proof draft written

TPM1 / TPO1
Terminal measure and pressure

T08–T09: iterated photon operators, sectors and the exact bounded-Borel measure identity.

T10–T11: two-sided partitions and the matter-partition defect between the two reference tori.

One proof manuscript and its companion outline

Current manuscript: 24 pages
Difficulty: 4/5

Terminal density and partition comparison.

T09: terminal measure

T11: pressure

finite Gaussian, trace and chessboard bounds

free coefficients and positive probe

Parent and component hypotheses for the closed-field covariance proof in QEM1

parent recovery and components

Q4R: response

all inputs

MODEL-SPECIFIC Proof draft written

VTM1 / VTO1
Volume transfer and component cumulants

All-volume translated-product bounds, the finite-periodic moment passage for each prescribed volume-first parent, and positive component fourth cumulants under the component Gaussian bound.

Proof manuscript and companion outline written. QEO1/T17–T18, B5, A3; numerical hypotheses retained.

Draft: 18 pages
Difficulty: 4/5

Volume losses and component passage.

structural inputs and conditional component cumulants

joint application and parameter checks

MODEL-SPECIFIC Proof draft written

QEM1 / QEO1 – Final QED3 construction

Shared cutoff conventions, common parameter admission (T16), the closed-field rotation proof, supported-tensor OS application (M18), and final assembly on one common full-measure component set. The transfer and component-cumulant estimates are imported from VTM1.

Target: subsequential field-strength limits whose almost-every ergodic component satisfies the full tensor OS axioms and has a strictly positive fourth cumulant.

Completed integration; full covariance proof retained locally. Earlier 263-page and 41-page editions are preserved.

Current manuscript: 32 pages
Difficulty: 5/5

Full closed-field rotation and tensor extension.

Proof draft written | Proof draft or integration revision in progress | Proof draft not written

Document units. Each box names its manuscript and companion outline, or the registered outline for a future manuscript. A companion outline is navigation for the same work, not a second proof task. This map follows the current QEM1 proof, which retains the closed-field covariance argument. DPM1 remains an alternative companion; it is not an input to this assembly. Inventory-only documents are not additional prerequisites. The nine renormalization and response units are shown individually in this map.

References. Labels such as T06, T09 and M18 identify statements in the full QEO1 outline. **Arrows.** These are uses of results inside documents, not a requirement to finish one entire paper before starting another. RRM1's early T06 identity feeds TPM1; TPM1's T09 measure feeds RRM1's later response proof. Joined inputs are required together; crossings without a dot do not join. VTM1 owns the finite-periodic moment application. QEM1 admits the common parameters before applying the conditional conclusions.

Status and length, 20 October 2026. Color concerns proof manuscripts; outline status is stated separately. Green records a written draft under its imports, not proof certification. Lengths are measured from the current linked PDFs. All displayed proof units have written drafts. QEM1 imports VTM1 and retains its finite-periodic, parameter, closed-field rotation and tensor arguments. The earlier comprehensive and integration editions are historical.

Series length: 336 pages in the current linked manuscripts. Each manuscript is counted once; outlines, alternative routes and historical editions are excluded.

New focused outlines: PRO1: positive probe; RRO1: reference response; VTO1: volume transfer and components; ECO1: continuum components and clustering.

Difficulty: 1 = routine/setup; 2 = standard techniques; 3 = substantial; 4 = hard; 5 = most demanding. Ratings concern the work inside each document, assuming its imported results. Reassessed against the current proofs; completion does not lower the rating.